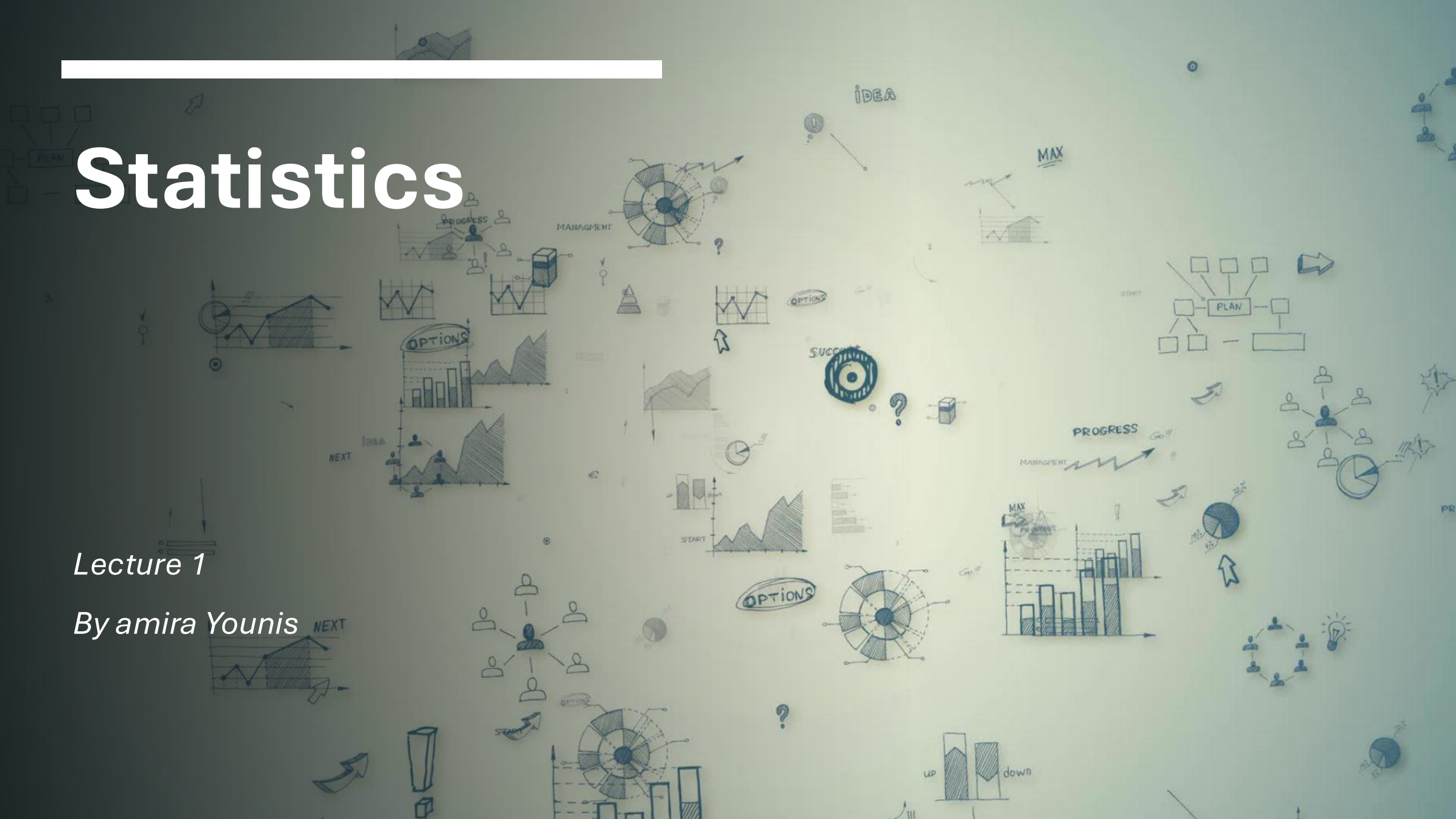


By amira Younis



statistics

statistics is the science of data.

- It involves Collecting, Classifying, Summarizing, Organizing, analyzing, and Interpreting numerical information.
- Statistics is used in several different disciplines (both scientific and non scientific) to make decisions and draw conclusions based on data.

How to collect data ?



Population

vs

sample

Population



The entire set of items or individuals of interest in a study. Denoted By N



Hard to define



Hard to observe

Sample



A subset selected from the larger population; Denoted by n .



less cost



Less time

How to choose sample

Randomness

Representative

Randomness

- Randomness in sampling means that every individual in the population has an equal chance of being selected. The selection is free from bias and not influenced by the researcher's preferences.
- **Imagine you have a jar with 100 marbles: 50 red and 50 blue.**

If you close your eyes and pick 10 marbles without looking, that's **random sampling**.

But if you only pick the 10 marbles from the top of the jar, that's **not random**, because the top may not represent the whole jar.

Representativeness

- A sample is **representative** if it accurately reflects the characteristics of the population from which it is drawn. The idea is that the sample should be a “mini version” of the population.
- **Suppose a university has 60% female and 40% male students.**

If you randomly choose 100 students and your sample has 58 females and 42 males, this is a **representative sample**, since the proportions are close to the real population.

But if you choose 100 students only from the engineering faculty (where most are male), your sample may have 80% males. This is **not representative** of the whole university.

quize



-
- A sample is called *random* if:
 - a) It has the same size as the population.
 - b) Every individual in the population has an equal chance of being selected.
 - c) It has exactly the same characteristics as the population.
 - d) It contains at least 30 individuals.

A university has 70% female and 30% male students. You take a random sample of 50 students and find 35 females and 15 males. Is your sample representative? Why or why not?

Types of data

```
graph TD; A[Types of data] --> B[Categorical]; A --> C[Numerical]; C --> D[Discrete]; C --> E[Continuous];
```

Categorical

Categorical data represents groups or categories.

Examples:

1. Car brands: Audi, BMW and Mercedes.
2. Answers to yes/no questions: yes and no

Numerical

Discrete

Continuous

Numerical data represents numbers. It is divided into two groups: discrete and continuous. Discrete data can be usually counted in a finite matter, while continuous is infinite and impossible to count.

Examples:

Discrete: # children you want to have, SAT score

Continuous: weight, height

Levels of measurement

Qualitative

Nominal

Ordinal

There are two qualitative levels: nominal and ordinal. The nominal level represents categories that cannot be put in any order, while ordinal represents categories that **can** be ordered.

Examples:

Nominal: four seasons (winter, spring, summer, autumn)

Ordinal: rating your meal (disgusting, unappetizing, neutral, tasty, and delicious)

Quantitative

Interval

Ratio

There are two quantitative levels: interval and ratio. They both represent “numbers”, however, ratios **have a true zero**, while intervals don’t.

Examples:

Interval: degrees Celsius and Fahrenheit

Ratio: degrees Kelvin, length

Visualize
categorical data



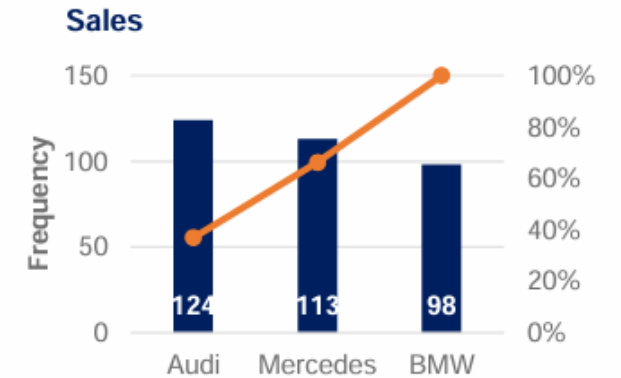
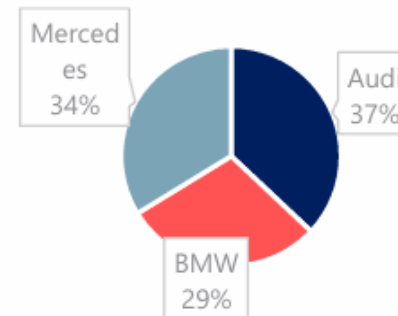
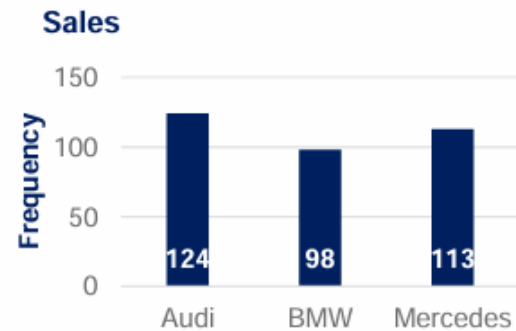
Frequency
distribution tables

Bar charts

Pie charts

Pareto diagrams

	Frequency
Audi	124
BMW	98
Mercedes	113
Total	335



Numerical
categorical data



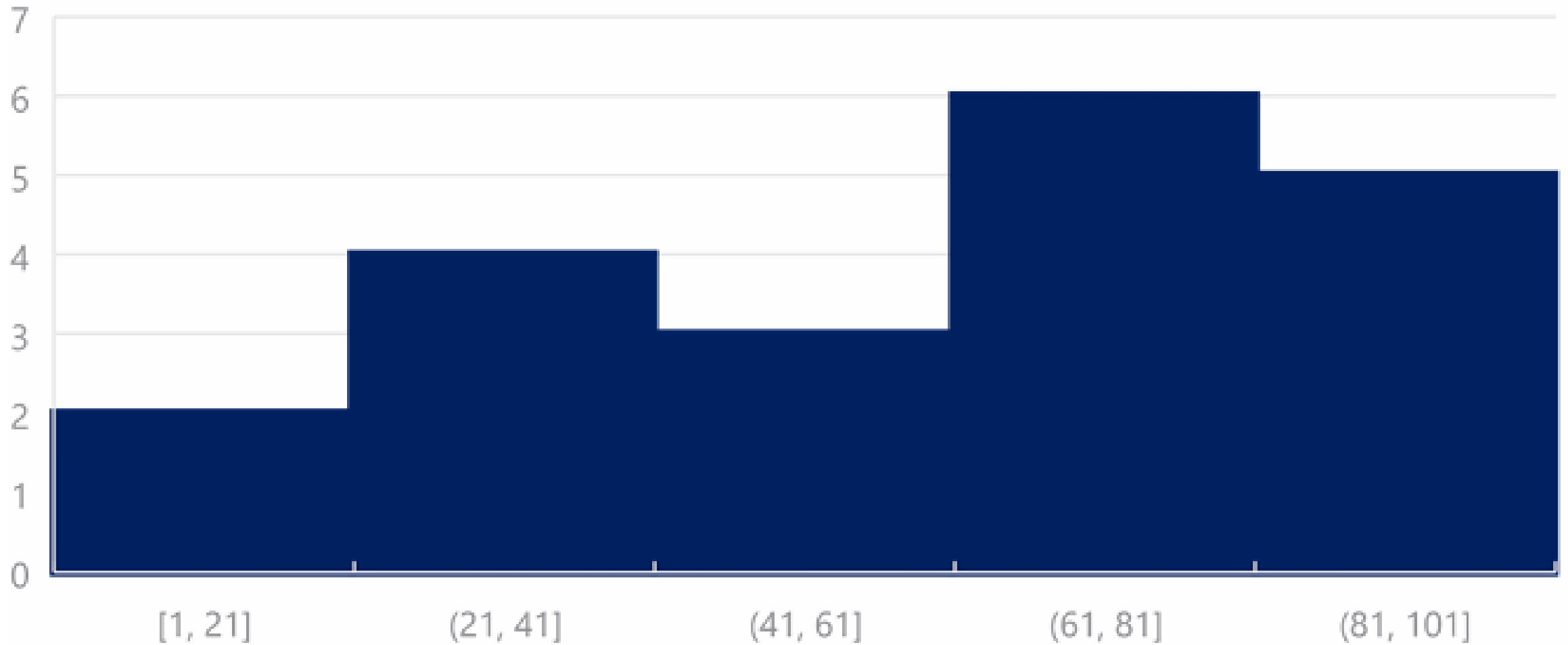
Frequency distribution table

Interval start	Interval end	Frequency	Relative frequency
1	21	2	0.10
21	41	4	0.20
41	61	3	0.15
61	81	6	0.30
81	101	5	0.25

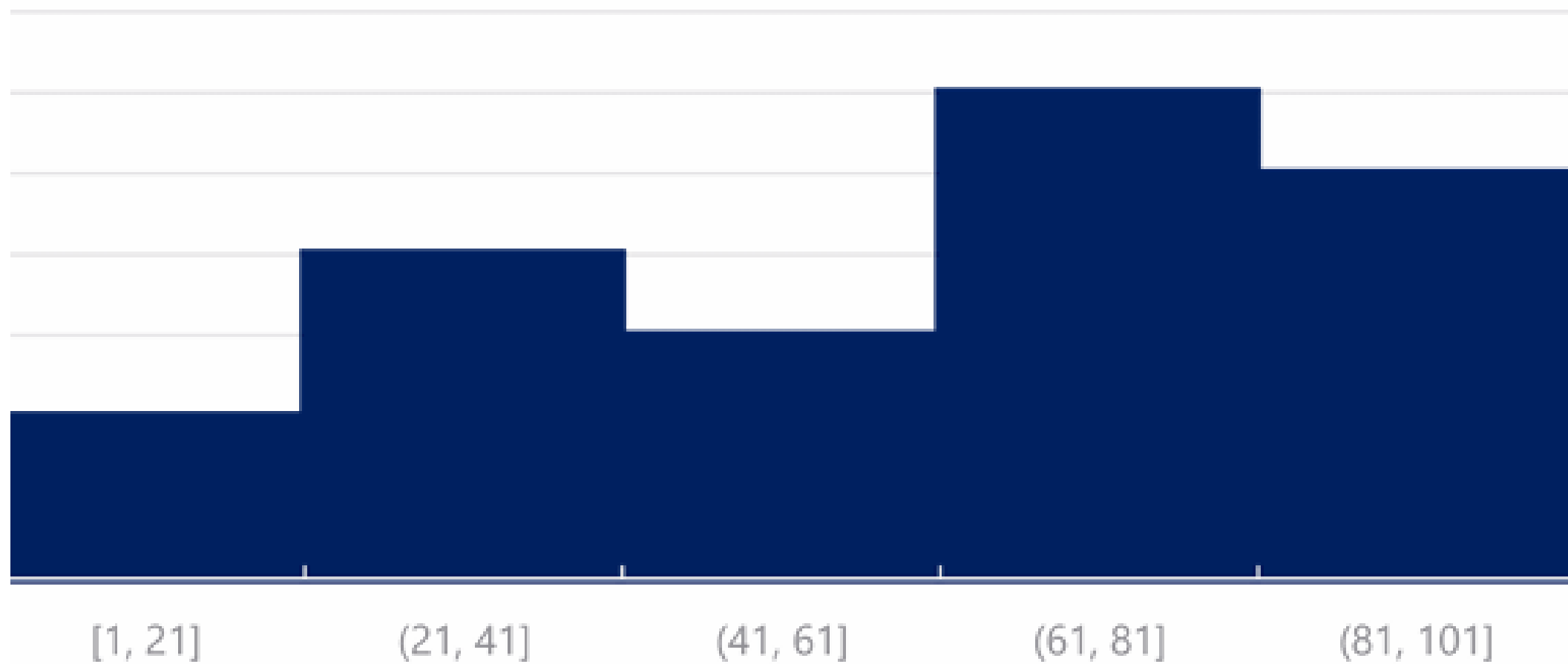
The interval width is calculated using the following formula:

$$\text{Interval width} = \frac{\text{Largest number} - \text{smallest number}}{\text{Number of desired intervals}}$$

Histogram



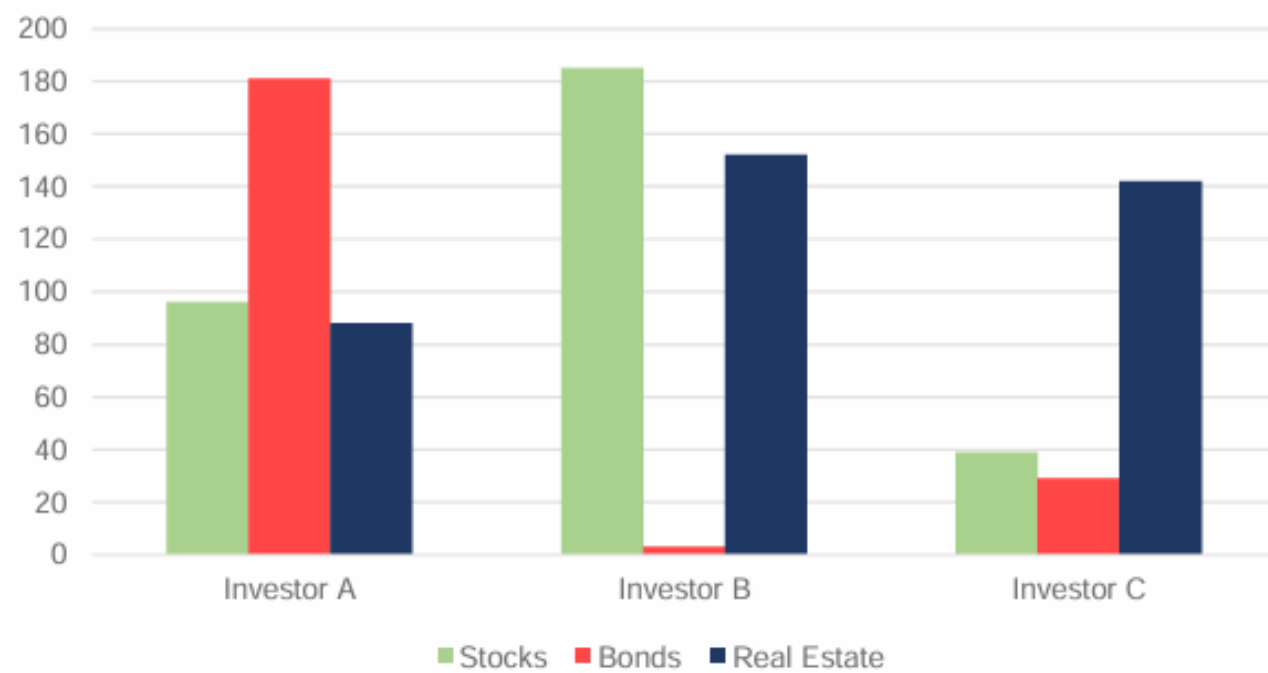
Histogram relative frequency

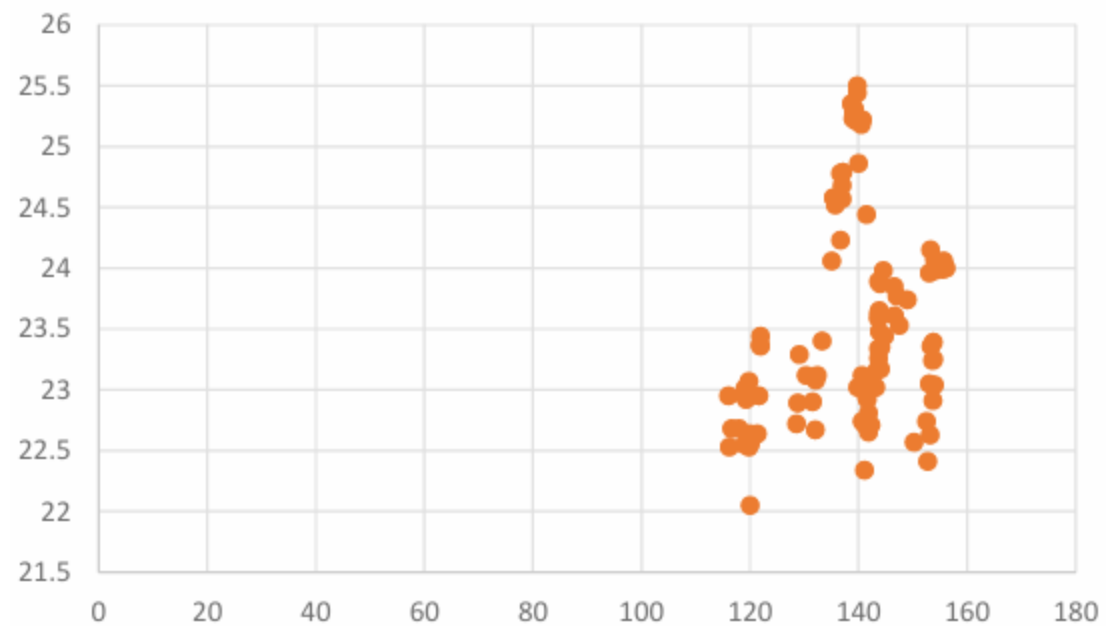
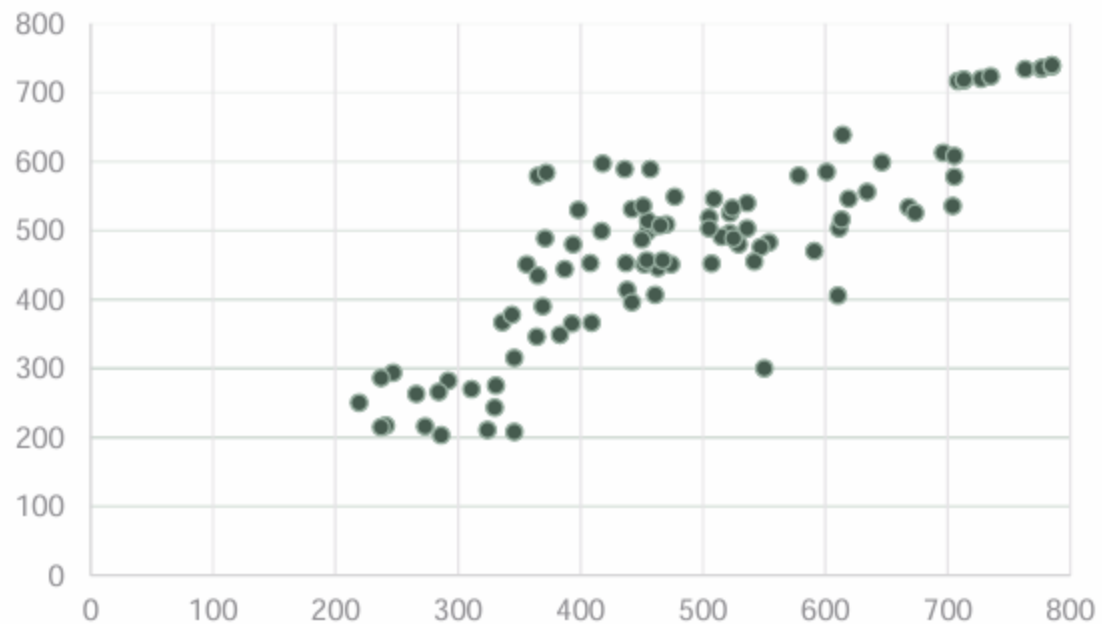


Type of investment \ Investor	Investor A	Investor B	Investor C	Total
Stocks	96	185	39	320
Bonds	181	3	29	213
Real Estate	88	152	142	382
Total	365	340	210	915

Type of investment \ Investor	Investor A	Investor B	Investor C	Total
Stocks	0.10	0.20	0.04	0.35
Bonds	0.20	0.00	0.03	0.23
Real Estate	0.10	0.17	0.16	0.42
Total	0.40	0.37	0.23	1.00

Side-by-side bar chart





Central tendencies

- A measure of central tendency is a single value that attempts to describe a set of data by identifying the central position within that set of data. As such, measures of central tendency are sometimes called measures of central location.

Staff	1	2	3	4	5	6	7	8	9	10
Salary	15k	18k	16k	14k	15k	15k	12k	17k	90k	95k

Mean , median ,mode

Mean

The mean is the most widely spread measure of central tendency. It is the simple average of the dataset.

Note: easily affected by outliers

The formula to calculate the mean is:

$$\frac{\sum_{i=1}^N x_i}{N} \quad \text{or}$$

$$\frac{x_1 + x_2 + x_3 + \dots + x_{N-1} + x_N}{N}$$

 In Excel, the mean is calculated by:

=AVERAGE()

Median

The median is the midpoint of the ordered dataset. It is not as popular as the mean, but is often used in academia and data science. That is since it is not affected by outliers.

In an ordered dataset, the median is the number at position $\frac{n+1}{2}$.

If this position is not a whole number, it, the median is the simple average of the two numbers at positions closest to the calculated value.

 In Excel, the median is calculated by:

=MEDIAN()

Mode

The mode is the value that occurs most often. A dataset can have 0 modes, 1 mode or multiple modes.

The mode is calculated simply by finding the value with the highest frequency.

 In Excel, the mode is calculated by:

=MODE.SNGL() -> returns one mode

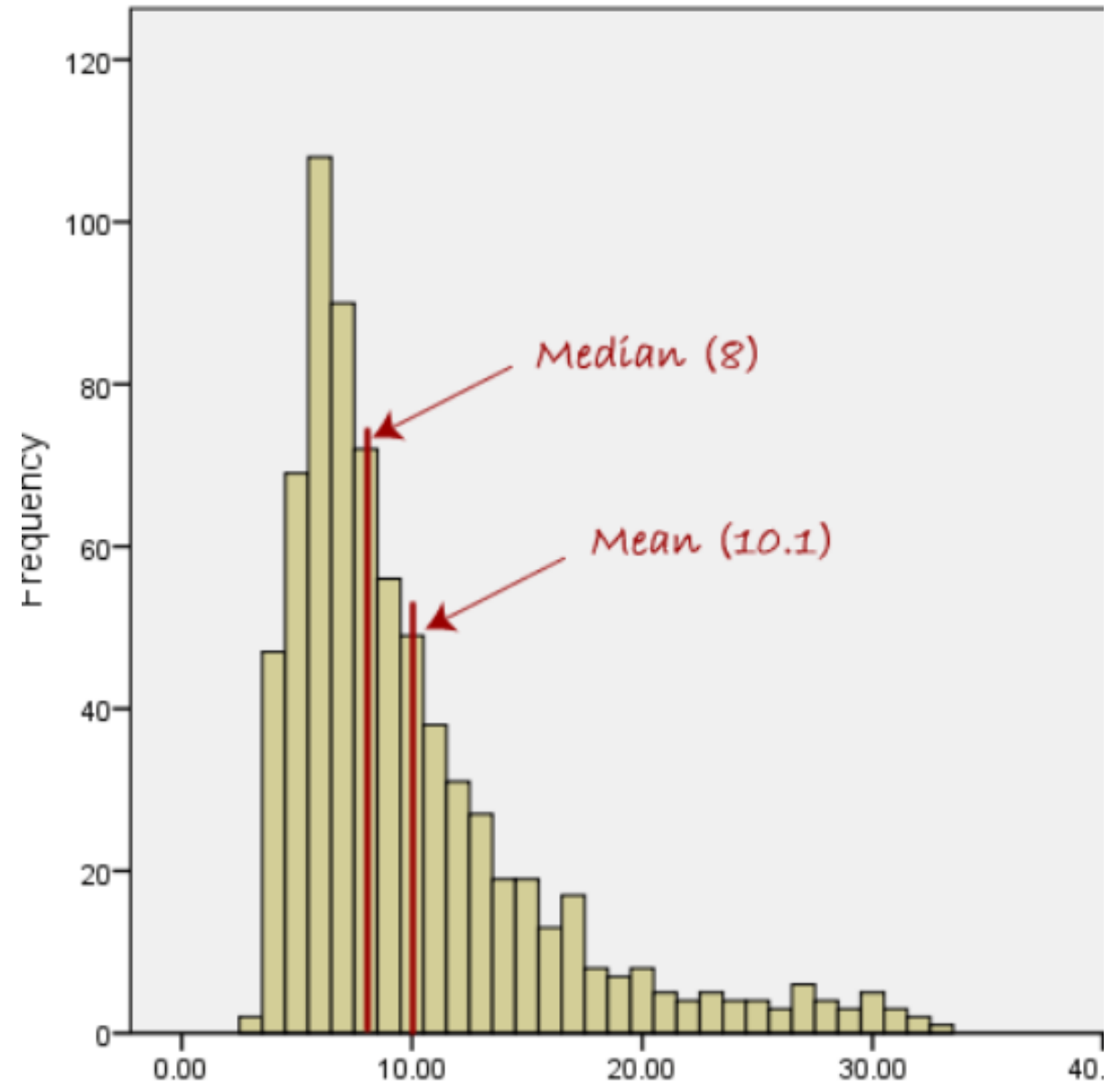
=MODE.MULT() -> returns an array with the modes. It is used when we have more than 1 mode.

Skewness

- Skewness is a measure of asymmetry that indicates whether the observations in a dataset are concentrated on one side. Right (positive) skewness looks like the one in the graph. It means that the outliers are to the right (long tail to the right). Left (negative) skewness means that the outliers are to the left

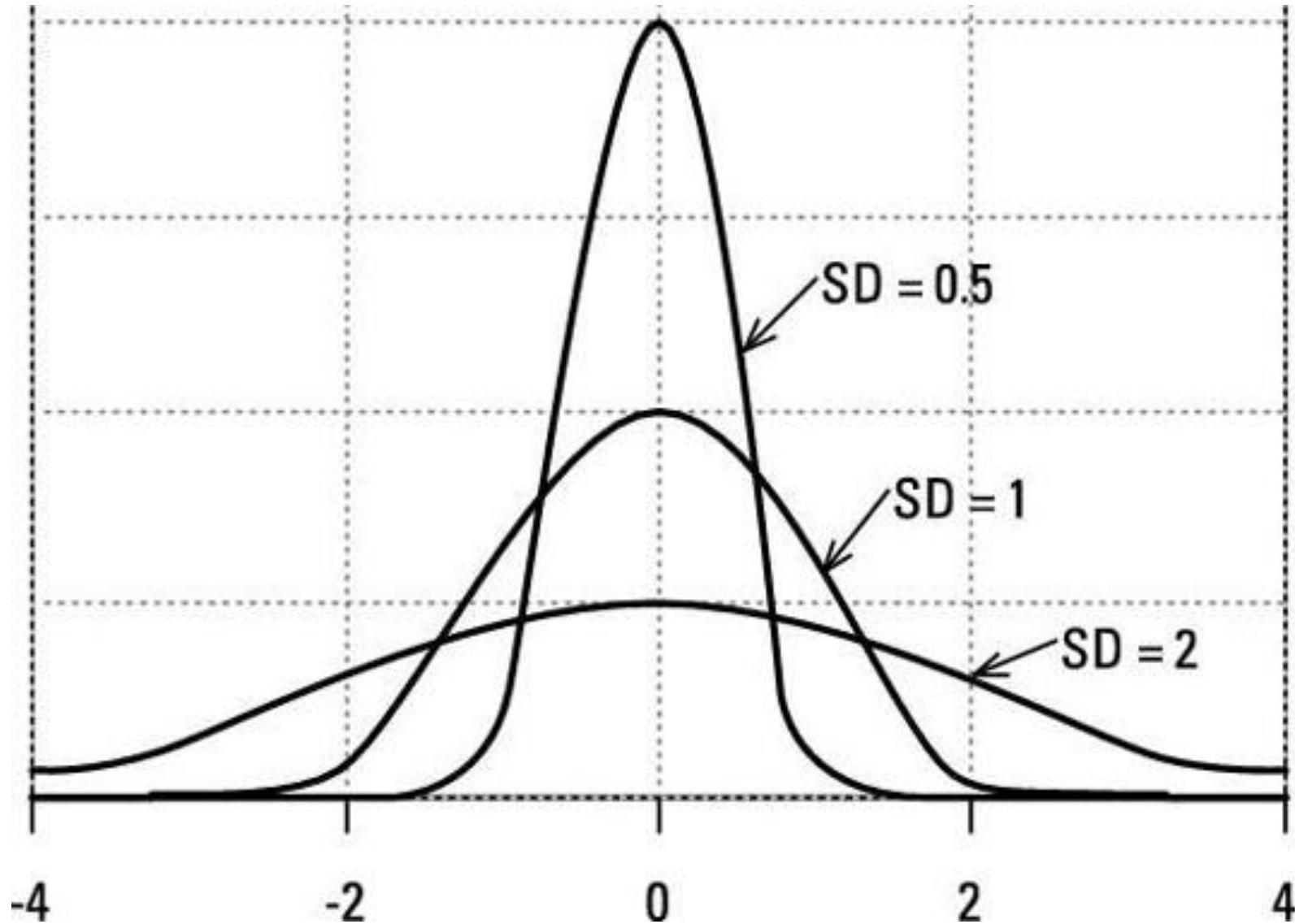
Formula to calculate skewness:

$$\frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^3}{\sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}}^3$$



.....

Type of Variable	Best measure of central tendency
Nominal	Mode
Ordinal	Median
Interval/Ratio (not skewed)	Mean
Interval/Ratio (skewed)	Median



Variance and standard deviation

Variance and standard deviation measure the dispersion of a set of data points around its mean value

Sample variance formula:

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$$

Population variance formula:

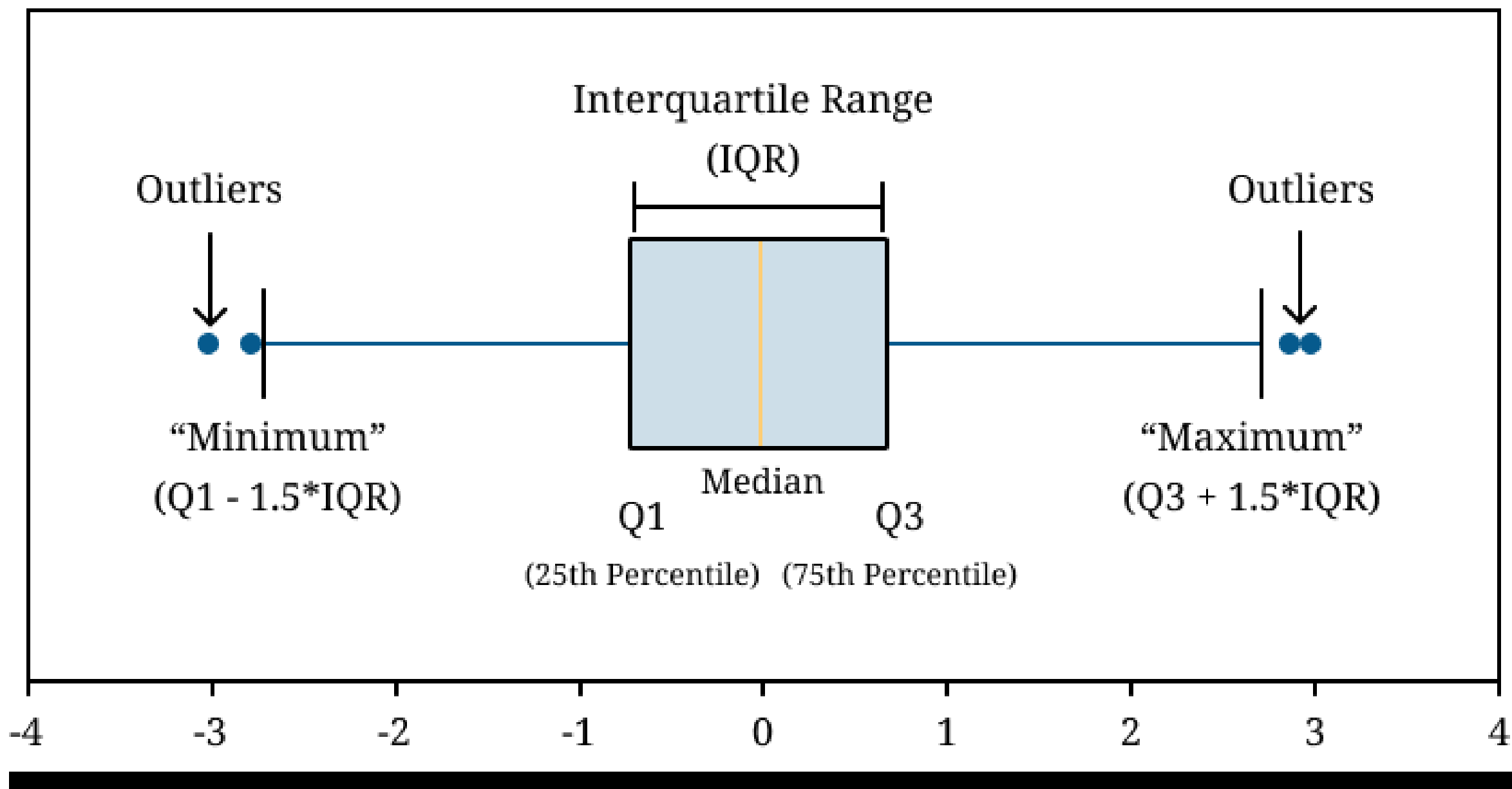
$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$

Sample standard deviation formula:

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}$$

Population standard deviation formula:

$$\sigma = \sqrt{\frac{\sum_{i=1}^N (x_i - \mu)^2}{N}}$$



Covariance and correlation

Covariance

Covariance is a measure of the joint variability of two variables.

- A positive covariance means that the two variables move together.
- A covariance of 0 means that the two variables are independent.
- A negative covariance means that the two variables move in opposite directions.

Covariance can take on values from $-\infty$ to $+\infty$. This is a problem as it is very hard to put such numbers into perspective.

Sample covariance formula:
$$s_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x}) * (y_i - \bar{y})}{n-1}$$

Population covariance formula:
$$\sigma_{xy} = \frac{\sum_{i=1}^N (x_i - \mu_x) * (y_i - \mu_y)}{N}$$

 In Excel, the covariance is calculated by:

Sample covariance: `=COVARIANCE.S()`

Correlation

Correlation is a measure of the joint variability of two variables. Unlike covariance, correlation could be thought of as a standardized measure. It takes on values between -1 and 1, thus it is easy for us to interpret the result.

- A correlation of 1, known as perfect positive correlation, means that one variable is perfectly explained by the other.
- A correlation of 0 means that the variables are independent.
- A correlation of -1, known as perfect negative correlation, means that one variable is explaining the other one perfectly, but they move in opposite directions.

Sample correlation formula:
$$r = \frac{s_{xy}}{s_x s_y}$$

Population correlation formula:
$$\rho = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$$

 In Excel, correlation is calculated by:

`=CORREL()`